

Trigonometry

ANSWER KEY



The unit circle and radian measure

- 1 Convert 210° to radians.
 $210 \times \frac{\pi}{180} = \frac{7\pi}{6}$.
 - 2 Convert $\frac{5\pi}{4}$ radians to degrees.
 $\frac{5\pi}{4} \times \frac{180}{\pi} = 225^\circ$.
 - 3 Find the exact value of $\sin(240^\circ)$.
 240° is in Quadrant III with reference angle 60° , where sine is negative: $\sin(240^\circ) = -\frac{\sqrt{3}}{2}$.
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Trigonometric identities

- 4 Simplify $\frac{1 - \sin^2 \theta}{\cos^2 \theta}$.
 $\frac{\cos^2 \theta}{\cos^2 \theta} = \cos \theta$.
 - 5 Simplify $\tan \theta \cdot \cos \theta$.
 $\frac{\sin \theta}{\cos \theta} \cdot \cos \theta = \sin \theta$.
 - 6 If $\sin \theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos \theta$ and $\tan \theta$.
 $\cos \theta = -\sqrt{1 - \frac{9}{25}} = -\frac{4}{5}$ (negative in Quadrant II). $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$.
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Double angle formulas

- 7 Given $\cos \theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.
 $\sin \theta = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$. $\sin 2\theta = 2\left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}$. $\cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{288}{169} = -\frac{119}{169}$.
 - 8 Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin \theta$ and $\cos \theta$ only.
 $\sin 4\theta = 2\sin 2\theta \cos 2\theta = 2(2\sin \theta \cos \theta)(1 - 2\sin^2 \theta) = 4\sin \theta \cos \theta (1 - 2\sin^2 \theta)$.
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Solving trigonometric equations

- 9 Solve $2\sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
 $\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$ or $\theta = 150^\circ$.
- 10 Solve $2\cos^2 \theta + \cos \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
Factoring: $(2\cos \theta - 1)(\cos \theta + 1) = 0$. $\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ, 300^\circ$. $\cos \theta = -1 \Rightarrow \theta = 180^\circ$.
Solutions: $60^\circ, 180^\circ, 300^\circ$.

Graphs of trigonometric functions

- 11 The graph shows $y = 3\sin(2x)$. State the amplitude and period.
Amplitude = 3. Period = $\frac{2\pi}{2} = \pi$.
- 12 The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?
Amplitude = 4. Period = $\frac{2\pi}{3}$. Writing $3x - \pi = 3\left(x - \frac{\pi}{3}\right)$, the phase shift is $\frac{\pi}{3}$ to the right.
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The Law of Sines

- 13 In triangle ABC , $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b , to the nearest hundredth.
 $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{10\sin 65^\circ}{\sin 40^\circ} \approx \frac{9.063}{0.643} \approx 14.10$.
- 14 In triangle ABC , $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.
 $h = b\sin A = 10\sin 35^\circ \approx 5.74$. Since $h < a < b$, two triangles are possible.
 $\sin B = \frac{b\sin A}{a} \approx 0.717 \Rightarrow B \approx 45.8^\circ$ or $B \approx 180^\circ - 45.8^\circ = 134.2^\circ$. Both are valid since $A + B < 180^\circ$ in each case.
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The Law of Cosines and the sine area formula

- 15 In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c , to the nearest hundredth, and the area of the triangle, to the nearest tenth.
 $c^2 = 11^2 + 14^2 - 2(11)(14)\cos 55^\circ \approx 317 - 176.66 = 140.34$, so $c \approx 11.85$. Area
 $= \frac{1}{2}(11)(14)\sin 55^\circ = 77\sin 55^\circ \approx 63.1$.
- 16 State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a , b , c and included angle C .
Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$. Area = $\frac{1}{2}ab\sin C$.
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Inverse trigonometric functions

- 17 Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.
 $\frac{\pi}{4}$.
- 18 Evaluate $\arccos(-1)$ exactly, in radians.
 π .

- 19 Find $\theta = \arctan(1)$ in degrees (principal value).
 45° .
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A well-designed word problem: locating a ship from two lighthouses

- 20 This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.

(a) $\angle ACB = 180^\circ - 72^\circ - 58^\circ = 50^\circ$. (b) AC is opposite $\angle B$: $AC = \frac{AB \sin B}{\sin C} = \frac{15 \sin 58^\circ}{\sin 50^\circ} \approx 16.6$ km. (c) BC is opposite $\angle A$: $BC = \frac{AB \sin A}{\sin C} = \frac{15 \sin 72^\circ}{\sin 50^\circ} \approx 18.6$ km.